THE RECIPROCAL-DEGREE CONSTANTS C_u OF THE THORP-SHUFFLE TRANSFER (OPENAI/MATH FAMILY 238): EXACT VALUES, AND THE EXPONENT OF ε_n

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Abstract. Family 238 of the openai/math repository proves optimal-order mixing of the Thorp shuffle. Its coset-to-Fourier transfer has the constant $C_u = \sup_{m\ge1}\sum_{\lambda\vdash m} f_\lambda^{-u}$, where f_λ is the degree of the Specht module. The family proves only $C_u < \infty$. We determine C_u exactly for $u \in \{1, \frac12, \frac{3}{10}, \frac15, \frac{3}{20}\}$: the supremum is attained at $m = 6, 8, 14, 22, 30$, and $C_1 = 473/144$. The proofs are computer-assisted, and each is carried out twice by two different methods. We also note that the main paper's $\varepsilon_n = O(n^{-5/2})$ is in fact $\Theta(n^{-e^*})$ with $e^* = 99\log_2(32/31) - \frac32 = 3.03456\ldots$. Both results are bookkeeping: they fill in constants that the original arguments leave implicit and add no new technique.

1. Statement

For $\lambda \vdash m$ let f_λ be the number of standard Young tableaux of shape λ, and put

$$Z_u(m) = \sum_{\lambda\vdash m} f_\lambda^{-u}, \qquad C_u = \sup_{m\ge1} Z_u(m) \qquad (u > 0).$$

This is the constant of [2, degrees.tex, eq. (l:degree-all-powers-equation); introduction.tex l. 14]. It is also the constant C_1 of [1, main.tex l. 420–424, Lemma `lem:reciprocal-degrees`], and the constant C_σ of remark R5 of this collection. Liebeck–Shalev [3] and Gamburd [4] show $Z_u(m) \to 2$, so $C_u < \infty$, but no value is given there or in family 238.

Theorem 1. *The supremum C_u is attained, at the m^* below, and*

u	1	1/2	3/10	1/5	3/20
m^*	6	8	14	22	30
$C_u = Z_u(m^*)$	$\frac{473}{144}$	5.90614654…	13.07321699…	34.58702880…	89.33317328…

Each value is enclosed in an interval of width $< 10^{-10}$; see §4. In addition, $C_{1/10} \ge Z_{1/10}(50) > 562.0737$.

Corollary 2. *In* [1, Prop. 6.1, eq. (h-12)], $A = \sqrt{32C_1} = \sqrt{946}/3 = 10.2524\ldots$. *In the transfer bounds $\|\hat f(\rho)\|_{\mathrm{HS}}^2 \le bBC_uD^{-1+(u+2)/b}$ of* [2, Thm. 1.1]*, the displayed values of C_u may be substituted for $u \in \{1, \frac12, \frac{3}{10}, \frac15, \frac{3}{20}\}$.*

Remark 3 (Growth as $u \to 0$). We have $\sum_\lambda f_\lambda^2 = m!$, so $f_\lambda \le \sqrt{m!}$ and $C_u \ge \max_m p(m)(m!)^{-u/2}$. Using $\log p(m) \sim \pi\sqrt{2m/3}$ and Stirling's formula, this gives $\log C_u \ge (\pi^2/6 + o(1))/(u\log(1/u))$. Hence C_u is not polynomial in $1/u$. We claim no matching upper bound.

Proposition 4 (ε_n)**.** *The quantity $\varepsilon_n = \frac{n^{3/2}}{2}\sqrt{(31/32)^{198d} + 32e^{-n/256}}$, $n = 2^d$, of* [1, main.tex l. 381–389, eq. (h-8)] *satisfies, for every $d \ge 15$,*

$$\tfrac12 n^{-e^*} \le \varepsilon_n \le \tfrac{\sqrt2}{2} n^{-e^*}, \qquad e^* = 99\log_2(32/31) - \tfrac32 \in (3.03456, 3.03457).$$

The paper states $O(n^{-5/2})$. That bound comes from replacing $198\log_2(32/31) = 9.0691\ldots$ by 8 (l. 401).

Date: October 9, 2026.

Proof. We have $(31/32)^{198d} = n^{-198\log_2(32/31)}$. For $d \geq 15$ the second term under the root is the smaller one; this is certified for $15 \leq d \leq 199$, and beyond that $n/256$ grows faster than d. The bounds on e^* are certified twice: by interval arithmetic, and by exact integer comparisons $2^{495q-p} \gtrless 31^{99q}$ (scripts `eps_n_exponent.py` and `independent/indep_eps.py`). □

The sharper exponent has no effect on the theorems of [1], which only use $\varepsilon_n \to 0$.

2. Route A (primary computation)

Lemma 5 (Ballot peeling)**.** *Let $\lambda = (L, \mu) \vdash m$, $k = m - L$ and $j = \min(k, L)$. Then $f_\lambda \geq f_\mu f_{(L,j)}$. Moreover $f_{(L,j)} = \binom{m}{k}\frac{m-2k+1}{m-k+1}$ if $k \leq L$, and $f_{(L,j)} = \mathrm{Cat}(L)$ if $k \geq L$.*

Sketch. Merge a standard tableau T of μ with a word w in L letters U and k letters D. The i-th D receives the label of the box of T that contains i. If the i-th D follows the $\min(i, L)$-th U, the result is standard for every T, because the box containing i lies in column $\leq \min(i, L)$. Different pairs (T, w) give different tableaux. The admissible words are counted by the ballot numbers. □

Route A uses the following inputs:

- exact enumeration for $m \leq 55$, with the hook formula and MPFI intervals;
- a three-index recursion over (size, first row, number of rows) for $55 < m \leq M_E$, using Lemma 5 and conjugation symmetry, in IEEE double arithmetic inflated by $1 + 2^{-40}$ per stage;
- a one-dimensional recursion $Z_u(m) \leq 2\big(1 + \sum_{L \geq \lceil\sqrt{m}\rceil} f_{(L,\min(k,L))}^{-u} R_u(m - L, L)\big)$ up to M_1;
- an analytic tail for $m > M_1$: the terms are bounded by $Z_u(k)$ for $k \leq K$, by $C\big((m+1)/\binom{m}{K+1}\big)^u$ for $K < k \leq m/2$ (at most $m/2$ terms; Lemma 5 gives $f_{(L,k)} \geq \binom{m}{k}/(k+1)$), and by a geometric series in $\mathrm{Cat}(L)$ for $k > m/2$.

The parameters (M_E, M_1, K) are $(120, 400, \lceil 2/u\rceil)$ for $u \geq \frac{1}{5}$ and $(200, 1000, 14)$ for $u = \frac{3}{20}$. Every majorant for $m > 55$ is below $Z_u(m^*)$; the largest is 69.59 at $m = 56$, for $u = \frac{3}{20}$. The output is in `scripts/expected_output/cu_bound.txt`.

3. Route B (independent re-verification)

Route B was written separately and shares no code with route A. It also uses a different peeling inequality and a different tail. The overall architecture (exact region, conjugation-symmetric three-index recursion, oriented one-dimensional recursion, tail) is the same elementary one as in route A.

Lemma 6 (First-row identity and bound)**.** *Let $\lambda = (L, \mu) \vdash m$, $k = m - L \geq 1$, $b = \mu_1$, and let μ'_j be the column lengths of μ. Then*

$$f_\lambda = \binom{m}{k}\frac{f_\mu}{\Pi}, \qquad \Pi = \prod_{j=1}^{b}\Big(1 + \frac{\mu'_j}{L - j + 1}\Big),$$

and

$$\Pi \leq \Big(1 + \frac{k(H_L - H_{L-b})}{b^2}\Big)^b \leq \min\Big\{\Big(\frac{m}{L}\Big)^L,\ \exp\frac{k}{L - \min(k, L) + 1}\Big\}.$$

Proof. Identity. The hooks of the boxes below the first row are the hooks of μ. The box $(1, j)$ has hook $L - j + 1 + \mu'_j$, where $\mu'_j = 0$ for $j > b$. So the hook formula gives $f_\lambda = \frac{m!}{k!} f_\mu / \prod_{j \leq L}(L - j + 1 + \mu'_j)$, and this product equals $L!\,\Pi$.

First bound. By AM–GM, $\Pi \leq \big(1 + \frac{1}{b}\sum_j \mu'_j w_j\big)^b$ with $w_j = 1/(L - j + 1)$. The sequence (μ'_j) is nonincreasing and (w_j) is nondecreasing, so Chebyshev's sum inequality gives $\sum_j \mu'_j w_j \leq \frac{1}{b}\big(\sum_j \mu'_j\big)\big(\sum_{j \leq b} w_j\big) = \frac{k}{b}(H_L - H_{L-b})$.

Second bound. Since $H_L - H_{L-b} \le b/(L-b+1)$, the middle term is at most $(1+\frac{k}{b(L-b+1)})^b$.

- This is at most $e^{k/(L-b+1)}$, and $b \le \min(k, L)$.
- It is also at most $(1+k/b)^b$. The function $t \mapsto t\log(1+k/t)$ is increasing, so this is at most $(1+k/L)^L = (m/L)^L$. □

Write $E_u(k,a,c)$ for the sum of f^{-u} over $\lambda \vdash k$ with $\lambda_1 = a$ and $\ell(\lambda) = c$. Then $E_u(k,a,c) = E_u(k,c,a)$. For $a \ge c \ge 2$, Lemma 6 gives

$$E_u(k,a,c) \le \binom{k}{a}^{-u} \sum_{b \le \min(a,k-a)} \Big(1 + \frac{(k-a)(H_a - H_{a-b})}{b^2}\Big)^{ub} E_u(k-a, b, c-1). \tag{1}$$

Orient λ so that $L = \lambda_1 \ge \ell(\lambda)$; then $L \ge \lceil\sqrt{m}\rceil$. This gives

$$Z_u(m) \le 2\Big(1 + \sum_{L=\lceil\sqrt{m}\rceil}^{m-1} \binom{m}{L}^{-u} \Pi^*(m,L)^u \, R_u(m-L, L)\Big). \tag{2}$$

Here Π^* is the minimum in Lemma 6, and $R_u(k,L) = \sum\{f_\mu^{-u} : \mu \vdash k,\ \mu_1 \le L,\ \ell(\mu) \le L-1\} \le Z_u(k)$.

Lemma 7 (Tail, route B)**.** *Let $u > 0$ and $0 < \tau < 2/3$. Let M, K be integers with $u(K+1) \ge 1 + 1/M$ and $3K \le M$. Put*

$$\kappa = 1 - \tau/2 - \tau^2/2, \qquad c_\tau = \min\{(1-\tau)\log\tfrac{1}{1-\tau},\ \tfrac{1}{3}\log 3\}, \qquad s_m = \lceil\sqrt{m}\rceil,$$
$$h(x) = (8x)^{u/2} e^{-u\kappa x}, \qquad \rho_s = (1+1/s)^{u/2} e^{-u\kappa}.$$

Assume $\rho_{s_M} < 1$ and $M \ge (1+u/2)/(uc_\tau)$. For $m \ge M$ let

$$T(m) = 2\Big[1 + \sum_{k=1}^{K} \binom{m}{k}^{-u} e^{\frac{uk}{m-2k+1}} Z_u(k) + Ce^u \frac{m}{3}\binom{m}{K+1}^{-u} + C\Big(\frac{h(s_m)}{1-\rho_{s_m}} + m(8m)^{u/2} e^{-uc_\tau m}\Big)\Big].$$

If $Z_u(k) \le C$ for all $k < M$ and $T(M) \le C$, then $Z_u(m) \le T(M)$ for all $m \ge M$.

Proof. We induct on m. Apply (2) with $R_u \le Z_u(k) \le C$, where $k = m - L$, and split into three ranges.

(a) $k \le K$. Here $k \le m/3$, so $k \le L$. Lemma 6 gives $\Pi \le e^{k/(m-2k+1)}$.

(b) $K < k \le m/3$. Here $m - 2k + 1 > k$, so $\Pi \le e$. Also $\binom{m}{k} \ge \binom{m}{K+1}$, and there are at most $m/3$ such terms.

(c) $k > m/3$, i.e. $s_m \le L < 2m/3$. Lemma 6 gives $\Pi \le (m/L)^L$. Robbins' bound $\binom{m}{L} \ge (8L(m-L)/m)^{-1/2} e^{mH(L/m)}$ [5, Ch. 10, Lemma 7] then gives $\binom{m}{L}(L/m)^L \ge (8L)^{-1/2} e^{g(L)}$, where $g(L) = (m-L)\log\frac{m}{m-L}$.

- For $L \le \tau m$: from $-\log(1-t) \ge t + t^2/2$ we get $g(L) \ge \kappa L$. The terms are $\le Ch(L)$, and $h(L+1)/h(L) \le \rho_{s_m}$, so their sum is at most $Ch(s_m)/(1-\rho_{s_m})$.
- For $\tau m < L < 2m/3$: g/m is concave in $t = L/m$, so $g \ge c_\tau m$. There are fewer than m terms.

Hence $Z_u(m) \le T(m)$.

It remains to show that T is nonincreasing in $m \ge M$. Range (a) is clear. Range (b): the ratio of consecutive values is at most $\exp(1/m - u(K+1)/(m+1)) \le 1$. Range (c), first piece: it is monotone in s_m, and h is decreasing. Range (c), second piece: the logarithmic derivative is $(1+u/2)/m - uc_\tau \le 0$. So $Z_u(m) \le T(m) \le T(M) \le C$. □

4. Certificates

Route B arithmetic.

- *Exact region $m \le 50$.* We enumerate the partitions and compute f_λ by the branching rule. The counts are checked against Euler's recurrence for $p(m)$, and $\sum f^2 = m!$ is checked.
- For $u = p/q$, each f^{-u} is enclosed between exact integer q-th roots $\lfloor (2^{80q}/f^p)^{1/q} \rfloor$ and $\lceil \cdot \rceil$, divided by 2^{80}. The enclosures are summed as integers. No floating point and no MPFR are used here.
- *Recursions.* (1) is run up to M_E and (2) up to M_1. Arithmetic is IEEE double; every weight and every stage is inflated by $1 + 2^{-30}$ (ln of the factor $\approx 9.3 \cdot 10^{-10}$). The tables of $\ln n!$ and H_n are computed at 40 digits, so the remaining error in a log-weight is a few ulp of numbers ≤ 700, i.e. $\lesssim 5 \cdot 10^{-13}$, and the margin is $> 10^3$. This step assumes IEEE-754 arithmetic and NumPy/libm `exp`, `log`, `log1p` accurate to a few ulp; it is not a formal interval computation (route A's weights are MPFI). A random sample of 2188 weights was recomputed in `mpmath.iv`, with 0 violations.
- *Tail.* Lemma 7 is evaluated in `mpmath.iv`. The constant C is $Z_u(m^*)$ rounded up at the 4th decimal.

Results.

u	max bound, $m > 55$ (A) value	at m	max bound, $m > 50$ (B) value	at m	$T(M_1)$ (A)	$T(M_1)$ (B)
1	2.0393	56	2.0436	51	2.055	2.004
1/2	2.4283	56	2.4627	51	2.281	2.094
3/10	3.8710	56	4.0540	51	3.084	2.647
1/5	11.562	56	11.753	51	10.322	4.726
3/20	69.593	56	52.277	51	10.83	26.75

Route B parameters (M_E, M_1) are $(120, 600)$, $(120, 600)$, $(150, 600)$, $(250, 1000)$, $(300, 1000)$. In the tail the script takes the best of $\tau \in \{\frac{1}{20}, \frac{1}{10}, \frac{1}{5}, \frac{3}{10}\}$ and $K \le 50$; it selects $\tau = \frac{1}{10}$ throughout and $K = 17, 44, 50, 50, 50$. In both routes every bound for $m \ne m^*$ is strictly below the lower endpoint of the enclosure of $Z_u(m^*)$, and the two routes give identical enclosures of $Z_u(m^*)$.

Remark 8 (Not covered). For $u = 1/10$, route A gives $C_{1/10} \le 1267.27$ (with $M_E = 450$, $M_1 = 2500$). This upper bound is not sharp and was not re-verified by route B: the unconstrained one-dimensional step loses too much at $M_E \le 400$. We do not claim it. Only the lower bound $C_{1/10} > 562.0737$ is confirmed by both routes.

5. Context and triviality

These constants were computed for an internal project, "H-1": full-domain CCA security of swap-or-not [6] in $O(\log N + \lambda)$ fixed rounds, obtained via the transfer of [2]. An internal audit showed that the asymptotic H-1 statement follows in a few lines from the family's own corollary ("$7n/8$-label error $o(1) \Rightarrow \mu * \mu \to U$"), HMR's Theorem 3, and submultiplicativity of total variation. The explicit constants improve that trivial route only by a factor of 3–5. We therefore make no claim about swap-or-not here. What remains is a computation of constants that the family leaves as "finite".

References

[1] OpenAI, *Optimal-order mixing of the Thorp shuffle*, openai/math family 238, `https://github.com/openai/math`, commit `fd4aeeb2ee4f`, 2026.

[2] OpenAI, *From partial permutation information to Fourier bounds*, openai/math family 238, same commit.

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Acknowledgement. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction. Independent adversarial reviews were run by separate AI sessions; they are not human peer review.